Determinant Divisibility of Centered Latin Squares: A Unified Theorem and the $\mathbb{F}_2$-Rank Obstruction

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Abstract

Let L be an $n \times n$ Latin square with entries in $\{1, \dots, n\}$ and let $E_{\text{std}} = P^T E P$ denote the Gram-projected $(n-1) \times (n-1)$ matrix of the centered operator $E = L - \frac{n+1}{2} J$ on the standard subspace V_{std}, where P is the basis matrix of V_{std}.

We prove three main results. First, $\frac{n^2}{\gcd(n,2)}$ divides $\det(E_{\text{std}})$ for every Latin square of every order $n \geq 2$; for odd n the stronger bound $n^2 \mid \det(E_{\text{std}})$ holds universally (*unified divisibility*). Second, for even n we give a complete $\mathbb{F}_2$-rank characterization of the Latin squares that achieve the improved divisibility $n^2 \mid \det(E_{\text{std}})$: when $n \equiv 2 \pmod 4$ the criterion is $\operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$; when $n \equiv 0 \pmod 4$ the criterion is $\operatorname{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ over $\mathbb{F}_2$. In both cases the parity pattern $L \bmod 2$ is a complete invariant (*complete binary criterion*). Equivalently, the obstruction to n^2-divisibility is entirely 2-adic: every odd prime factor of n contributes its share unconditionally. Third, we prove that counterexamples to n^2-divisibility exist for every $n \equiv 2 \pmod 4$, $n \geq 6$, via an explicit infinite family of skip-one circulant parity patterns (*universal counterexamples*).

These results are complemented by p-adic lower bounds: $v_p(\det A) \geq v_p(n) + \max(0, k_p - 1)$ for every odd prime $p \mid n$, and $v_2(\det A) \geq v_2(n/2) + \max(0, k_2 - 1)$ when n is even (empirically tight across sampled instances up to $n \leq 20$), a cokernel computation for cyclic Latin squares, and exhaustive/constructive counterexample analyses at orders 6 and 10. The determinant side of the story may be viewed as complementary to earlier work on determinants and permanents of Latin squares [7, 3].

Keywords: Latin squares, determinant divisibility, $\mathbb{F}_2$-rank, centered matrix, constant-weight code, parity pattern.

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1 Introduction

A *Latin square* of order n is an $n \times n$ array $L = (L_{ij})$ whose entries belong to $\{1, \dots, n\}$ and in which every row and every column is a permutation of $\{1, \dots, n\}$. Latin squares are fundamental objects in combinatorics with deep connections to algebra, coding theory, and experimental design [5, 14]. Determinant questions for Latin squares have also been studied from other viewpoints, including order-specific determinant calculations and determinant / permanent polynomials [7, 3].

Given a Latin square L, the *centered matrix* $E = L - \frac{n+1}{2} J$ (where J is the all-ones matrix) satisfies $E\mathbf{1} = \mathbf{0}$ and $\mathbf{1}^T E = \mathbf{0}$, so $\operatorname{rank}(E) \leq n - 1$ and E acts naturally on the hyperplane $V_{\text{std}} = \{x \in \mathbb{R}^n : \sum x_i = 0\}$. The Gram-projected matrix $E_{\text{std}} = P^T E P$ (where P is the basis matrix of V_{std}; see Section 2) carries algebraic information about the Latin square that is invisible to purely combinatorial invariants.

In this paper we study the arithmetic of $\det(E_{\text{std}})$. The main contributions are threefold.

(a) *Unified divisibility.* For every Latin square of order $n \geq 2$,

$$\frac{n^2}{\gcd(n,2)} \,\Big|\, \det(E_{\text{std}}).$$

In particular, if n is odd then the stronger divisibility $n^2 \mid \det(E_{\mathrm{std}})$ holds universally (Theorem 3.3).

(b) *Complete binary criterion.* The obstruction to n^2-divisibility is entirely 2-adic: every odd prime factor of n contributes its share unconditionally. When $n \equiv 2 \pmod 4$, the criterion is rank-theoretic:

$$n^2 \mid \det(E_{\mathrm{std}}) \iff \mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1.$$

When $n \equiv 0 \pmod 4$, rank alone is not sufficient; the complete criterion is the adjugate condition

$$\mathrm{adj}(A \bmod 2)\cdot \mathbf{1} = \mathbf{0} \quad \text{in } \mathbb{F}_2^{n-1}.$$

In both congruence classes, the parity pattern $L \bmod 2$ is a complete invariant for the n^2-divisibility question.

(c) *Universal counterexamples.* For every $n \equiv 2 \pmod 4$, $n \geq 6$, there exist Latin squares violating n^2-divisibility (Theorem 4.14). Concrete examples at $n = 6$ (exhaustive) and $n = 10$ (constructed) are given in Section 7.

Secondary results include the constant row-weight theorem for $A \bmod 2$ (Theorem 5.1), the balanced realizability proposition (Theorem 4.12), the sharp p-adic bounds via Smith normal form (Theorems 6.1 and 6.3), and the cokernel computation for cyclic Latin squares (Theorem 3.9).

The proof rests on two independent mechanisms. The first is a geometric factor of n arising from the Gram matrix of the standard basis for V_{std}. The second is an arithmetic factor: the row sums of A are always divisible by n (when n is odd) or by $n/2$ (when n is even), which forces $\det(A)$ to acquire a corresponding factor via the adjugate identity. These two factors combine to yield the quadratic bound.

The paper is organized as follows. Section 2 fixes notation and establishes the Gram factorization. Section 3 proves the unified divisibility theorem. Section 4 establishes the binary criterion: the rank characterization for $n \equiv 2 \pmod 4$, the adjugate criterion for $n \equiv 0 \pmod 4$, and the universality of counterexamples for all $n \equiv 2 \pmod 4$. Section 5 derives structural properties of $A \bmod 2$. Section 6 develops the Smith-normal-form and p-adic bounds. Section 7 presents the counterexamples. Section 8 summarizes the computational methods. Section 9 concludes with a summary, related directions, and open problems.

2 Preliminaries and notation

Throughout, L is an $n \times n$ Latin square with entries in $\{1, \ldots, n\}$, indexed by $0 \leq i, j \leq n-1$. For a prime p and a nonzero integer m, we write $v_p(m)$ for the p-adic valuation; by convention, $v_p(0) = +\infty$. All p-adic lower bounds in this paper are therefore vacuously satisfied when $\det(A) = 0$ (which can occur, e.g. at $n = 4$).

Definition 2.1 (Centered matrix). Set $m = \frac{n+1}{2}$ and $E = L - mJ$. The entries of E are half-integers when n is even and integers when n is odd.

Since every row and column of L sums to $\frac{n(n+1)}{2} = nm$, we have $E\mathbf{1} = \mathbf{0}$ and $\mathbf{1}^T E = \mathbf{0}$.

Definition 2.2 (Standard representation). Let $V_{\mathrm{std}} = \ker(\mathbf{1}^T) \subset \mathbb{R}^n$ and define the basis $b_i = e_i - e_{n-1}$ for $0 \leq i \leq n-2$, where e_i is the i-th standard unit vector. The *Gram matrix* is

$$G = P^T P, \qquad P = [b_0 \mid \cdots \mid b_{n-2}] \in \mathbb{R}^{n\times(n-1)},$$

so $G = I_{n-1} + J_{n-1}$ and $\det(G) = n$.

Definition 2.3 (Difference matrix)**.** Define the $(n-1)\times(n-1)$ integer matrix

$$A_{ij} = L_{ij} - L_{i,n-1}, \qquad 0 \le i,j \le n-2.$$

Lemma 2.4 (Gram factorization)**.** $E_{\mathrm{std}} = G\cdot A$*, so* $\det(E_{\mathrm{std}}) = n\cdot\det(A)$.

Proof. Since $E\mathbf{1}=\mathbf{0}$, the operator E maps V_{std} into itself. Writing $EP = PA$, the first $n-1$ rows are verified entry-wise by

$$(EP)_{k,j} = E_{k,j} - E_{k,n-1} = L_{kj} - L_{k,n-1} = A_{kj}$$

for $0 \le k,j \le n-2$, since the m-terms cancel. For the last row, use $\mathbf{1}^T E = \mathbf{0}^T$. For each $0 \le j \le n-2$,

$$(EP)_{n-1,j} = E_{n-1,j} - E_{n-1,n-1} = -\sum_{r=0}^{n-2}(E_{rj} - E_{r,n-1}) = -\sum_{r=0}^{n-2} A_{rj}.$$

This is exactly the last row of PA, since the last row of P is $(-1,\dots,-1)$. Thus $EP = PA$, and so we obtain $E_{\mathrm{std}} = P^T EP = P^T PA = GA$, whence $\det(E_{\mathrm{std}}) = \det(G)\det(A) = n\det(A)$. □

Remark 2.5. The factorization $E_{\mathrm{std}} = GA$ distinguishes two objects:

- A is the coordinate matrix of $E|_{V_{\mathrm{std}}}$ in the basis $\{b_i\}$: it records the action $Eb_j = \sum_i A_{ij}\, b_i$.
- $E_{\mathrm{std}} = GA$ is the Gram-projected matrix, with entries $(E_{\mathrm{std}})_{ij} = \langle b_i, Eb_j\rangle = (P^T EP)_{ij}$.

In particular, $\det(E_{\mathrm{std}}) = n\cdot\det(A)$, where the factor $n = \det(G)$ reflects the non-orthonormality of $\{b_i\}$. The invariant $\det(A)\in\mathbb{Z}$ is the determinant of the *operator* $E|_{V_{\mathrm{std}}}$; $\det(E_{\mathrm{std}})$ absorbs an additional geometric factor. Both are studied in this paper; most divisibility statements are formulated for $\det(E_{\mathrm{std}})$ and then equivalently expressed in terms of $\det(A)$.

3 The unified divisibility theorem

Lemma 3.1 (Row-sum identity)**.** *For every* $0 \le i \le n-2$,

$$\sum_{j=0}^{n-2} A_{ij} \;=\; \frac{n}{2}\big(n+1-2L_{i,n-1}\big).$$

Write $c_i = n+1-2L_{i,n-1}$.

- *If* n *is even, then* c_i *is odd and* $A\mathbf{1} = \frac{n}{2}\,\mathbf{c}$.
- *If* n *is odd, then* c_i *is even; writing* $u_i = c_i/2 \in \mathbb{Z}$*, one has* $A\mathbf{1} = n\,\mathbf{u}$.

Proof. We compute

$$\begin{aligned}\sum_{j=0}^{n-2} A_{ij} &= \sum_{j=0}^{n-2}(L_{ij} - L_{i,n-1}) = \left(\sum_{j=0}^{n-1} L_{ij}\right) - L_{i,n-1} - (n-1)L_{i,n-1}\\ &= \frac{n(n+1)}{2} - nL_{i,n-1} = \frac{n}{2}(n+1-2L_{i,n-1}).\end{aligned}$$

Since $2L_{i,n-1}$ is always even: if n is even then $n+1$ is odd, so c_i is odd; if n is odd then $n+1$ is even, so c_i is even and $u_i = c_i/2 = (n+1-2L_{i,n-1})/2$ is an integer. □

Theorem 3.2 (Unified divisibility)**.** *For every Latin square* L *of order* $n \ge 2$,

$$\frac{n^2}{\gcd(n,2)} \;\Big|\; \det(E_{\mathrm{std}}).$$

Proof. By Theorem 2.4, $\det(E_{\mathrm{std}}) = n\det(A)$. It suffices to show that $\frac{n}{\gcd(n,2)} \mid \det(A)$. We treat odd and even n separately.

Case 1: n odd. By Theorem 3.1, every row sum of A is $\frac{n}{2}(n+1-2L_{i,n-1})$. Since $n+1$ is even when n is odd, $(n+1-2L_{i,n-1})/2$ is an integer, and so each row sum equals $n \cdot \left(\frac{n+1-2L_{i,n-1}}{2}\right)$. Hence $A\mathbf{1} \equiv \mathbf{0} \pmod{n}$. By the adjugate identity $\operatorname{adj}(A) \cdot A\mathbf{1} = \det(A) \cdot \mathbf{1}$, it follows that $n \mid \det(A)$. Combined with $\det(E_{\mathrm{std}}) = n\det(A)$, we obtain $n^2 \mid \det(E_{\mathrm{std}})$.

Case 2: n even. Set $h = n/2$. By Theorem 3.1, every row sum equals $h \cdot (\text{odd})$, so $A\mathbf{1} \equiv \mathbf{0} \pmod{h}$. By the adjugate identity, $h \mid \det(A)$. Therefore $\det(E_{\mathrm{std}}) = n\det(A)$ is divisible by $n \cdot h = n^2/2 = n^2/\gcd(n,2)$. ☐

Corollary 3.3. *If n is odd, then $n^2 \mid \det(E_{\mathrm{std}})$.*

Proof. This is Case 1 of Theorem 3.2: each row sum of A is divisible by n, so $n \mid \det(A)$ and $\det(E_{\mathrm{std}}) = n\det(A)$ is divisible by n^2. ☐

Remark 3.4 (Prime-by-prime decomposition). Let p be an odd prime dividing n. By Theorem 3.1, each row sum of A is divisible by n when n is odd (since $A\mathbf{1} = n\,\mathbf{u}$) and by $n/2$ when n is even (since $A\mathbf{1} = \frac{n}{2}\,\mathbf{c}$ with c_i odd). In either case $v_p(n) \le v_p(\text{row sum})$, so $A\mathbf{1} \equiv \mathbf{0} \pmod{p^{v_p(n)}}$ and therefore $p^{v_p(n)} \mid \det(A)$.

Proposition 3.5 (Sharpness)**.** *The bound in Theorem 3.2 is sharp for $n \in \{2, 6, 8, 10\}$. Specifically, there exist Latin squares satisfying $\frac{n^2}{2} \mid \det(E_{\mathrm{std}})$ but $n^2 \nmid \det(E_{\mathrm{std}})$ at each of these orders.*

(i) *For $n = 2$, $\det(E_{\mathrm{std}}) = -2$ and $4 \nmid -2$.*

(ii) *For $n = 6$, there are exactly 576 reduced Latin squares with $18 \mid \det(E_{\mathrm{std}})$ but $36 \nmid \det(E_{\mathrm{std}})$ (Theorem 7.1).*

(iii) *For $n = 8$, explicit Jacobson–Matthews samples satisfy $32 \mid \det(E_{\mathrm{std}})$ but $64 \nmid \det(E_{\mathrm{std}})$; all such squares have $\dim\ker_{\mathbb{F}_2}(A \bmod 2) = 1$, explained by the adjugate criterion (Theorem 4.7).*

(iv) *For $n = 10$, explicit Latin squares satisfy $50 \mid \det(E_{\mathrm{std}})$ but $100 \nmid \det(E_{\mathrm{std}})$ (Theorem 7.5).*

Remark 3.6. At $n = 4$, exhaustive computation over all 576 Latin squares shows that $n^2 = 16$ divides $\det(E_{\mathrm{std}})$ universally; the weaker bound $n^2/2 = 8$ is therefore *not* sharp at this order. Whether $n = 4$ is the only even order where the bound can be strengthened remains open.

Proof. Items (i), (ii), and (iv) are exhibited in Section 7. Item (iii): a sharpness witness at $n = 8$ is the Latin square obtained from the first Jacobson–Matthews sample (with $5n^2 = 320$ mixing steps, seed = 0) satisfying $\dim\ker_{\mathbb{F}_2} = 1$; its determinant is verified by Bareiss elimination to satisfy $32 \mid \det(E_{\mathrm{std}})$ but $64 \nmid \det(E_{\mathrm{std}})$. ☐

Remark 3.7. In a corpus of 10 000 Jacobson–Matthews samples at $n = 8$ ($5n^2$ mixing steps each), approximately 11% satisfy $32 \mid \det(E_{\mathrm{std}})$ but $64 \nmid \det(E_{\mathrm{std}})$. This sampling frequency is not claimed as a rigorous proportion of all Latin squares of order 8.

Proposition 3.8 (Cyclic formula)**.** *For the cyclic Latin square $L_{ij} = (i + j) \bmod n + 1$ of any order $n \ge 2$:*

$$\det(E_{\mathrm{std}})_{\mathrm{cyc}} = (-1)^{\lfloor n/2 \rfloor} \cdot n^{n-1}.$$

In particular, $|\det(E_{\mathrm{std}})| = n^{n-1}$, i.e. the exact exponent of n dividing $\det(E_{\mathrm{std}})$ is $n-1$. (Singular Latin squares with $\det(A) = 0$ exist already at $n = 4$, so no absolute optimality claim is intended here.)

Proof. The centered matrix E has entries $E_{ij} = (i{+}j) \bmod n - (n{-}1)/2 =: h\big((i{+}j) \bmod n\big)$ and is real symmetric since $E_{ij} = E_{ji}$. Because $E\mathbf{1} = \mathbf{0}$, the eigenvalues of A are all real (since $A = G^{-1}E_{\mathrm{std}}$ is similar to the real symmetric matrix $G^{-1/2}E_{\mathrm{std}}\,G^{-1/2}$), and $\det(A) = \prod \mu_k$.

Since E_{ij} depends only on $(i{+}j) \bmod n$ (a reverse-circulant or Hankel-circulant structure), the DFT vectors satisfy $Ev^{(k)} = \hat{h}(k)\,v^{(n-k)}$ for each $k = 1, \dots, n-1$, where $v_j^{(k)} = \omega^{jk}$, $\omega = e^{2\pi i/n}$, and $\hat{h}(k) = \sum_{j=0}^{n-1} h(j)\,\omega^{jk}$. (The reversal $v^{(k)} \mapsto v^{(n-k)}$ arises because $(Mv^{(k)})_i = \sum_j h(i{+}j)\omega^{jk} = \omega^{-ik}\hat{h}(k) = \hat{h}(k)\,v_i^{(n-k)}$.) Evaluating explicitly: $h(j) = j - (n{-}1)/2$ for $j = 0, \dots, n{-}1$ gives $\hat{h}(k) = n/(\omega^k{-}1)$ for $k \neq 0$, since $\sum_{j=0}^{n-1} j\,\omega^{jk} = -n/(\omega^k{-}1)$ and $\sum_j \omega^{jk} = 0$. Writing $\hat{h}(k) = a_k + ib_k$ and passing to the real basis $u_k = \Re\, v^{(k)}$, $w_k = \Im\, v^{(k)}$, one finds (see Davis [4], Chap. 3 for the general spectral theory of circulant and Hankel-circulant matrices) that E acts on $\langle u_k, w_k\rangle$ by the matrix $\left[\begin{smallmatrix} a_k & b_k \\ b_k & -a_k \end{smallmatrix}\right]$, whose eigenvalues are $\pm|\hat{h}(k)| = \pm n/|\omega^k{-}1|$—one positive and one negative. Since $v^{(n-k)} = \overline{v^{(k)}}$, the pair $\{k, n{-}k\}$ shares the same real subspace $W_k = \langle u_k, w_k\rangle$, yielding $\lfloor (n{-}1)/2 \rfloor$ two-dimensional blocks. When n is even, the real vector $v^{(n/2)} = (1, -1, 1, -1, \dots) \in V_{\mathrm{std}}$ is an eigenvector with eigenvalue $\hat{h}(n/2) = n/(\omega^{n/2}{-}1) = -n/2 < 0$.

Absolute value. By the evaluation identity $\prod_{k=1}^{n-1}(1{-}\omega^k) = n$ (substitute $x = 1$ in $(x^n{-}1)/(x{-}1)$),

$$|\det(A)| = \prod_{k=1}^{n-1} |\hat{h}(k)| = \frac{n^{n-1}}{\prod_{k=1}^{n-1} |\omega^k{-}1|} = \frac{n^{n-1}}{n} = n^{n-2}.$$

Sign. The number of negative eigenvalues is $\lfloor (n{-}1)/2 \rfloor$ (one per conjugate pair) plus 1 when n is even (from $\hat{h}(n/2) < 0$), totalling $\lfloor n/2 \rfloor$. Hence $\det(A) = (-1)^{\lfloor n/2 \rfloor} n^{n-2}$ and $\det(E_{\mathrm{std}}) = n\,\det(A) = (-1)^{\lfloor n/2 \rfloor} n^{n-1}$. □

The cyclic determinant formula yields a complete description of the Smith normal form of A (see [12] for general SNF theory) and, consequently, the *cokernel* $K(A) \cong \mathbb{Z}^{n-1}/\mathrm{Im}(A)$ (sometimes called the *critical group* by analogy with graph Laplacians [2]; we avoid the term "sandpile group" since A is not a Laplacian matrix).

Corollary 3.9 (Cokernel of cyclic Latin squares). *For the cyclic Latin square of any order $n \geq 2$, the reduced matrix A has Smith normal form* $\mathrm{diag}(1, n, n, \dots, n)$ *with $n-2$ copies of n. Equivalently,*

$$K(A)_{\mathrm{cyc}} \;\cong\; (\mathbb{Z}/n\mathbb{Z})^{n-2}.$$

Proof. Write $A = R + n\Delta$ where $R_{ij} = j + 1$ for all i, j (a rank-one integer matrix) and Δ is an integer matrix; this follows from $A_{ij} = (i{+}j) \bmod n - (i{+}n{-}1) \bmod n \equiv j + 1 \pmod{n}$.

Step 1 (Divisibility). By multilinearity of the determinant in rows, every $k \times k$ minor of A expands as a sum over subsets S of the k row indices, with each term carrying the factor $n^{|S|}$ from the Δ-rows. Since R has identical rows, any term with $|S| \leq k{-}2$ (i.e., ≥ 2 rows from R) vanishes. Hence every $k \times k$ minor is divisible by n^{k-1}, so the k-th determinantal divisor $\Delta_k :=$ gcd of all $k \times k$ minors satisfies $n^{k-1} \mid \Delta_k$.

Step 2 (Tightness). Since $\gcd(A_{0,0}, \dots, A_{0,n-2}) = \gcd(1{-}n, \dots, -1) = 1$, we have $\Delta_1 = 1$. By Theorem 3.8, $|\det A| = n^{n-2}$, so $\Delta_{n-1} = n^{n-2}$.

Step 3 (Pigeonhole). The invariant factors $d_k = \Delta_k/\Delta_{k-1}$ satisfy $d_1 = 1$ and $d_2 \mid d_3 \mid \cdots \mid d_{n-1}$ with $\prod_{k=2}^{n-1} d_k = n^{n-2}$. For $k = 2$, Step 1 gives $n \mid \Delta_2 = d_1 d_2 = d_2$, so $d_2 \geq n$. The product bound yields $d_2^{n-2} \leq n^{n-2}$, hence $d_2 = n$. Since $d_3 \mid \cdots \mid d_{n-1}$ with product n^{n-3} and each $d_k \geq d_2 = n$, the same squeeze gives $d_k = n$ for all $k \geq 2$ by induction. □

4 The binary criterion: rank and adjugate

In this section we restrict to *even* n and study when the stronger divisibility $n^2 \mid \det(E_{\mathrm{std}})$ holds. The characterization is complete for $n \equiv 2 \pmod 4$ via the $\mathbb{F}_2$-rank; for $n \equiv 0 \pmod 4$, a subtler adjugate criterion resolves the question.

Definition 4.1 (Parity pattern)**.** The *parity pattern* of L is the binary matrix $\mathcal{P} = L \bmod 2 \in \mathbb{F}_2^{n\times n}$. Since each row and column of L is a permutation of $\{1,\dots,n\}$ and n is even, each row and column of $\mathcal{P}$ contains exactly $n/2$ ones and $n/2$ zeros. We call such a matrix *doubly balanced.*

Lemma 4.2. *Over* $\mathbb{F}_2$*:*

$$(A \bmod 2)_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i,n-1}, \qquad 0 \le i,j \le n-2,$$

where $\oplus$ *denotes addition in* $\mathbb{F}_2$*. In particular,* $A \bmod 2$ *depends only on* $\mathcal{P}$*.*

Proof. $A_{ij} = L_{ij} - L_{i,n-1}$, and subtraction coincides with addition over $\mathbb{F}_2$. ◻

Theorem 4.3 ($\mathbb{F}_2$-rank characterization)**.** *Let* $n \equiv 2 \pmod 4$*. Then:*

$$n^2 \mid \det(E_{\mathrm{std}}) \iff \operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1.$$

Proof. By Theorem 2.4, $n^2 \mid \det(E_{\mathrm{std}})$ iff $n \mid \det(A)$. Write $n = 2q$ with q odd (since $v_2(n) = 1$).

Odd prime factors. For every odd prime $p \mid n$, $p^{v_p(n)} \mid \det(A)$ holds universally by Theorem 3.4. This accounts for the full odd part q of n.

The prime 2. It remains to show $2 \mid \det(A)$. We already know $h \mid \det(A)$ from Theorem 3.2, where $h = n/2 = q$ is odd. Thus $q \mid \det(A)$ and the question reduces to whether $2 \mid \det(A)$:

$$n \mid \det(A) \iff 2 \mid \det(A) \iff \det(A \bmod 2) = 0 \iff \operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1.$$

The second equivalence uses $\det(A) \bmod 2 = \det(A \bmod 2)$ over $\mathbb{F}_2$. ◻

Remark 4.4 ($n \equiv 0 \pmod 4$: the F$_2$-rank is necessary but not sufficient). When $n \equiv 0 \pmod 4$, $h = n/2$ is even. By Theorem 3.1, every row sum of A equals $h\cdot(\text{odd})$, which is even. Hence $A\mathbf{1} \equiv \mathbf{0} \pmod 2$, i.e. $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$, so $\operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) \le n-2$ and $2 \mid \det(A)$ unconditionally. However, for $n^2 \mid \det(E_{\mathrm{std}})$ one needs $2^a \mid \det(A)$ where $a = v_2(n) \ge 2$, and the rank condition provides only one factor of 2 beyond the 2^{a-1} already guaranteed by $h \mid \det(A)$. Indeed, for $n = 8$, approximately 11% of Jacobson–Matthews samples achieve $v_2(\det E_{\mathrm{std}}) = 5 = 2v_2(n)-1$, so that $\frac{n^2}{2} \mid \det(E_{\mathrm{std}})$ but $n^2 \nmid \det(E_{\mathrm{std}})$. A sufficient condition for $n^2 \mid \det(E_{\mathrm{std}})$ when $n \equiv 0 \pmod 4$ is $\dim\ker_{\mathbb{F}_2}(A \bmod 2) \ge 2$ (Theorem 5.7); a complete characterization is given by Theorem 4.7.

Corollary 4.5. *If* $n \equiv 2 \pmod 4$*, then* $\mathbf{1} \notin \ker_{\mathbb{F}_2}(A \bmod 2)$*. In particular, the kernel mechanism of Theorem 4.4 is absent, and whether* $n^2 \mid \det(E_{\mathrm{std}})$ *depends on the specific Latin square.*

Proof. When $n \equiv 2 \pmod 4$, $h = n/2$ is odd, so each row sum $h \cdot (\text{odd})$ is odd. Hence $(A \bmod 2)\mathbf{1} = \mathbf{1} \neq \mathbf{0}$ over $\mathbb{F}_2$. ◻

Remark 4.6. For $n \equiv 2 \pmod 4$, Theorem 4.3 together with Theorem 4.2 shows that the parity pattern $\mathcal{P} = L \bmod 2$ is a *complete invariant* for the n^2-divisibility question: two Latin squares sharing the same parity pattern either both satisfy $n^2 \mid \det(E_{\mathrm{std}})$, or neither does.

This was verified exhaustively at $n = 6$: among 1252 distinct parity patterns arising from the 9408 reduced Latin squares, every Latin square with a given pattern yields the same $\mathbb{F}_2$-rank for $A \bmod 2$, with zero exceptions.

We now resolve the case $n \equiv 0 \pmod 4$.

Theorem 4.7 (Adjugate criterion for $n \equiv 0 \pmod 4$)**.** *Let $n \equiv 0 \pmod 4$ and set $B = A \bmod 2 \in \mathbb{F}_2^{(n-1)\times(n-1)}$. Then*

$$n^2 \mid \det(E_{\mathrm{std}}) \iff \mathrm{adj}(B) \cdot \mathbf{1} = \mathbf{0} \quad \text{in } \mathbb{F}_2^{n-1}.$$

Proof. By Theorem 2.4, $n^2 \mid \det(E_{\mathrm{std}})$ iff $n \mid \det(A)$. Set $h = n/2$. By Theorem 3.2, $h \mid \det(A)$; write $\det(A) = hq$ with $q \in \mathbb{Z}$. Then $n \mid \det(A)$ iff $2 \mid q$.

Step 1. From the adjugate identity $\det(A) \cdot \mathbf{1} = \mathrm{adj}(A) \cdot A\mathbf{1}$ and $A\mathbf{1} = h\,\mathbf{c}$ (Theorem 3.1), where $c_i = n + 1 - 2L_{i,n-1}$ is always odd, we obtain

$$q \cdot \mathbf{1} = \mathrm{adj}(A) \cdot \mathbf{c}.$$

Step 2. Reduce modulo 2. Since every c_i is odd, $\mathbf{c} \equiv \mathbf{1} \pmod 2$. Since the determinant is polynomial in the matrix entries, $\mathrm{adj}(A) \bmod 2 = \mathrm{adj}(A \bmod 2) = \mathrm{adj}(B)$. Hence

$$q \bmod 2 \;=\; [\mathrm{adj}(B) \cdot \mathbf{1}]_i \qquad \text{for every } i,$$

and $2 \mid q$ iff $\mathrm{adj}(B) \cdot \mathbf{1} = \mathbf{0}$ in $\mathbb{F}_2^{n-1}$. □

Corollary 4.8 (Left-kernel parity criterion)**.** *Let $n \equiv 0 \pmod 4$ and $B = A \bmod 2$ with $\dim \ker_{\mathbb{F}_2}(B) = 1$. Let $\lambda \in \mathbb{F}_2^{n-1}$ generate $\ker(B^T)$. Then*

$$n^2 \mid \det(E_{\mathrm{std}}) \iff \mathrm{wt}(\lambda) \equiv 0 \pmod 2.$$

Proof. Since $\mathrm{rank}(B) = n-2$ and $B\mathbf{1} = \mathbf{0}$ (so $\ker(B) = \mathrm{span}\{\mathbf{1}\}$), every column of $\mathrm{adj}(B)$ lies in $\ker(B) = \mathrm{span}\{\mathbf{1}\}$ and every row in $\ker(B^T) = \mathrm{span}\{\lambda\}$. Since $\mathrm{adj}(B) \neq 0$ (at least one $(n-2) \times (n-2)$ minor is nonzero), we have $\mathrm{adj}(B) = \mathbf{1} \cdot \lambda^T$. Hence $\mathrm{adj}(B) \cdot \mathbf{1} = (\lambda^T \mathbf{1}) \cdot \mathbf{1}$, which vanishes iff $\mathrm{wt}(\lambda)$ is even. The result follows from Theorem 4.7. □

Remark 4.9 (Special cases of the adjugate criterion).

(i) If $\dim \ker_{\mathbb{F}_2}(B) \geq 2$, then $\mathrm{adj}(B) = 0$ and the criterion holds automatically, recovering Theorem 5.7.

(ii) As a consistency check, Theorem 4.8 was tested on 347 Jacobson–Matthews samples with $\dim\ker = 1$ at $n = 8, 12, 16$; no misclassifications were found.

Example 4.10. For a Latin square L of order 8 with rows $(4,3,6,8,5,7,2,1)$, $(2,7,4,1,6,8,3,5)$, $(8,5,3,4,1,6,7,2)$, $(5,1,7,3,2,4,6,8)$, $(3,8,2,6,7,5,1,4)$, $(7,6,8,2,4,1,5,3)$, $(6,2,1,5,8,3,4,7)$, $(1,4,5,7,3,2,8,6)$, the difference matrix $B = A \bmod 2 \in \mathbb{F}_2^{7\times 7}$ has $\mathrm{rank}_{\mathbb{F}_2}(B) = 6$, so $\dim \ker_{\mathbb{F}_2}(B) = 1$ and Theorem 4.7 applies. Computing $\mathrm{adj}(B) \cdot \mathbf{1} = \mathbf{1} \neq \mathbf{0}$, the criterion predicts $64 \nmid \det(E_{\mathrm{std}})$. Direct computation confirms: $\det(A) = -245{,}244$ with $v_2(\det(A)) = 2$, hence $v_2(\det(E_{\mathrm{std}})) = v_2(8 \cdot \det(A)) = 5 < 6 = v_2(64)$.

Corollary 4.11 (Parity pattern completeness)**.** *For every even n, the question $n^2 \mid \det(E_{\mathrm{std}})$ depends only on the parity pattern $\mathcal{P} = L \bmod 2$.*

Proof. For $n \equiv 2 \pmod 4$, this follows from Theorems 4.2 and 4.3. For $n \equiv 0 \pmod 4$, Theorem 4.7 involves only $B = A \bmod 2$, which depends on $\mathcal{P}$ by Theorem 4.2. □

4.1 Universal counterexamples in the 2 mod 4 case

The $\mathbb{F}_2$-rank characterization (Theorem 4.3) reduces the question of counterexample existence to finding doubly balanced binary matrices with full $\mathbb{F}_2$-rank. We now prove that such counterexamples exist for every $n \equiv 2 \pmod 4$, $n \geq 6$.

Proposition 4.12 (Balanced realizability)**.** *Let n be even. Every doubly balanced binary matrix $\mathcal{P} \in \mathbb{F}_2^{n\times n}$ is the parity pattern of some Latin square of order n.*

Proof. Since every row and column of $\mathcal{P}$ has exactly $n/2$ ones, the bipartite graph G with biadjacency matrix $\mathcal{P}$ is $(n/2)$-regular. By König's edge-coloring theorem [9], G decomposes into $n/2$ perfect matchings $M_1, \dots, M_{n/2}$. Assign to the edges of M_r the odd symbol $2r-1$ (for $r = 1, \dots, n/2$). Similarly, the complement $J - \mathcal{P}$ defines an $(n/2)$-regular bipartite graph that decomposes into $n/2$ perfect matchings $M'_1, \dots, M'_{n/2}$; assign to the edges of M'_s the even symbol $2s$ (for $s = 1, \dots, n/2$). The resulting $n \times n$ array L is a Latin square (each row and column receives exactly one label from each matching), and $L \bmod 2 = \mathcal{P}$. $\square$

Lemma 4.13 (Rank bridge). *Let $n \equiv 2 \pmod 4$, and let $\mathcal{P} \in \mathbb{F}_2^{n\times n}$ be doubly balanced. Define $B \in \mathbb{F}_2^{(n-1)\times(n-1)}$ by $B_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i,n-1}$ for $0 \le i, j \le n-2$. Then*

$$\operatorname{rank}_{\mathbb{F}_2}(B) = \operatorname{rank}_{\mathbb{F}_2}(\mathcal{P}) - 1.$$

Proof. Since $n/2$ is odd, every row and column sum of $\mathcal{P}$ equals $n/2 \equiv 1 \pmod 2$, so $\mathcal{P}\mathbf{1} = \mathbf{1}$ and $\mathbf{1}^T\mathcal{P} = \mathbf{1}^T$ over $\mathbb{F}_2$.

Let $E = \{y \in \mathbb{F}_2^n : \mathbf{1}^T y = 0\}$; then $\mathcal{P}$ maps E to E (since $\mathbf{1}^T(\mathcal{P}y) = \mathbf{1}^T y = 0$ for $y \in E$). Define $T\colon \mathbb{F}_2^{n-1} \to \mathbb{F}_2^n$ by $(T(x))_j = x_j$ for $j \le n-2$ and $(T(x))_{n-1} = \mathbf{1}^T x$ (i.e. the last coordinate is the parity of the first $n-1$). Then T is an isomorphism $\mathbb{F}_2^{n-1} \xrightarrow{\sim} E$. A direct computation shows that $T^{-1}\mathcal{P}\,T = B$ (the (i,j)-entry of $T^{-1}\mathcal{P}\,T$ is $\mathcal{P}_{ij} \oplus \mathcal{P}_{i,n-1} = B_{ij}$). Therefore $\operatorname{rank}(B) = \operatorname{rank}(\mathcal{P}|_E)$.

Since $\mathcal{P}\mathbf{1} = \mathbf{1} \ne \mathbf{0}$, the vector $\mathbf{1}$ is not in $\ker(\mathcal{P})$, so $\ker(\mathcal{P})$ is entirely contained in E. Thus $\dim\ker(\mathcal{P}|_E) = \dim\ker(\mathcal{P})$ and

$$\operatorname{rank}(B) = \dim E - \dim\ker(\mathcal{P}) = (n-1) - (n - \operatorname{rank}(\mathcal{P})) = \operatorname{rank}(\mathcal{P}) - 1. \qquad \square$$

Theorem 4.14 (Universal counterexamples for $n \equiv 2 \pmod 4$). *For every $n \equiv 2 \pmod 4$, $n \ge 6$, there exists a Latin square L of order n such that $n^2 \nmid \det(E_{\mathrm{std}})$.*

More precisely, let $n = 2k$ with $k \ge 3$ odd, and define

$$S_n = \{0, 1, \dots, k-2, k\} \subseteq \mathbb{Z}_n.$$

Then the circulant parity pattern $\mathcal{P}_{S_n}$ (with $(\mathcal{P}_{S_n})_{ij} = \mathbf{1}_{(j-i) \bmod n \in S_n}$) is doubly balanced, lifts to a Latin square by Theorem 4.12, and satisfies $\operatorname{rank}_{\mathbb{F}_2}(\mathcal{P}_{S_n}) = n$. By Theorem 4.13, $\operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) = n-1$, and by Theorem 4.3, $n^2 \nmid \det(E_{\mathrm{std}})$.

Proof. Since $|S_n| = k = n/2$, the circulant $\mathcal{P}_{S_n}$ has exactly $n/2$ ones in every row and column, so it is doubly balanced. By the standard circulant criterion over $\mathbb{F}_2$, $\mathcal{P}_{S_n}$ is invertible if and only if

$$\gcd\big(f_n(x),\, x^n - 1\big) = 1,$$

where $f_n(x) = \sum_{s\in S_n} x^s = 1 + x + \cdots + x^{k-2} + x^k$ is the indicator polynomial. Since $\operatorname{char}\mathbb{F}_2 = 2$ and $n = 2k$,

$$x^n - 1 = x^n + 1 = (x^k + 1)^2,$$

so it suffices to show $\gcd(f_n,\, x^k + 1) = 1$.

Suppose α is a root of $x^k + 1$ in the algebraic closure $\overline{\mathbb{F}}_2$, i.e. $\alpha^k = 1$.

Case $\alpha = 1$. $f_n(1) = |S_n| = k \equiv 1 \pmod 2$ (since k is odd), so $f_n(1) \ne 0$.

Case $\alpha \ne 1$. Observe that

$$(x+1)\, f_n(x) = 1 + x^{k-1} + x^k + x^{k+1},$$

which is verified by expanding $(x+1)(1 + x + \cdots + x^{k-2} + x^k)$ over $\mathbb{F}_2$. Evaluating at α with $\alpha^k = 1$:

$$(\alpha+1)\, f_n(\alpha) = 1 + \alpha^{k-1} + \alpha^k + \alpha^{k+1} = 1 + \alpha^{-1} + 1 + \alpha = \alpha^{-1} + \alpha.$$

If $f_n(\alpha) = 0$ then $\alpha^{-1} + \alpha = 0$, i.e. $\alpha^2 = 1$, whence $\alpha = 1$ (the only solution in characteristic 2), contradicting $\alpha \neq 1$.

Therefore $\gcd(f_n, x^k + 1) = 1$, so $\mathcal{P}_{S_n}$ is invertible over $\mathbb{F}_2$, i.e. $\mathrm{rank}_{\mathbb{F}_2}(\mathcal{P}_{S_n}) = n$. By Theorem 4.13, $\mathrm{rank}_{\mathbb{F}_2}(B) = n - 1$ where $B = A \bmod 2$ (Theorem 4.2). Theorem 4.3 then gives $n^2 \nmid \det(E_{\mathrm{std}})$. □

Remark 4.15. The skip-one family S_n was originally identified by exhaustive search over all cyclic subsets $S \subseteq \mathbb{Z}_n$ with $|S| = n/2$, and verified computationally to give full $\mathbb{F}_2$-rank for all $n \equiv 2 \pmod 4$ up to $n = 198$ (49 values). The algebraic proof above shows the family works for all such n.

Witness determinant values from one specific lift (via the perfect matching decomposition of Theorem 4.12) are non-canonical: different matching decompositions may produce different Latin squares with the same parity pattern, and hence different determinant values. However, the $\mathbb{F}_2$-rank—and therefore the counterexample property—depends only on the parity pattern (Theorem 4.11).

Remark 4.16 (Consecutive interval degeneration). The complementary choice $T_n = \{0, 1, \ldots, k-1\}$ (a consecutive interval) always yields $\mathrm{rank}_{\mathbb{F}_2}(\mathcal{P}_{T_n}) = n/2$, the minimum possible for a doubly balanced circulant. This is because the indicator polynomial $g(x) = 1 + x + \cdots + x^{k-1} = (x^k + 1)/(x + 1)$ divides $x^k + 1$ over $\mathbb{F}_2$, so $\gcd(g, x^{2k} + 1) = (x^k + 1)/(x + 1)$ has degree $k - 1$, and the rank drops by $k - 1$. The skip-one modification $S_n = T_n \setminus \{k - 1\} \cup \{k\}$ breaks this divisibility and restores full rank.

5 Structural properties of A mod 2

In this section we derive finer structural results about the binary matrix $A \bmod 2$.

Theorem 5.1 (Constant row weight). *Let n be even and let $\mathcal{P} = L \bmod 2$ be doubly balanced (i.e. every row and column of $\mathcal{P}$ has exactly $n/2$ ones). Then every row of $A \bmod 2$ has Hamming weight exactly $n/2$.*

Proof. By Theorem 4.2, $(A \bmod 2)_{ij} = \mathcal{P}_{ij} \oplus \mathcal{P}_{i,n-1}$ for $0 \leq j \leq n - 2$. Fix a row i. Since $\mathcal{P}$ is doubly balanced, row i of $\mathcal{P}$ has exactly $n/2$ ones among its n entries.

Case $\mathcal{P}_{i,n-1} = 1$. Then $(A \bmod 2)_{ij} = 1$ iff $\mathcal{P}_{ij} = 0$. Among the first $n - 1$ columns, the number of zeros is $(n - 1) - (n/2 - 1) = n/2$ (subtracting the $n/2 - 1$ ones from columns $0, \ldots, n - 2$ given that one of the $n/2$ total ones is in column $n - 1$). Hence $\mathrm{wt}(\text{row } i) = n/2$.

Case $\mathcal{P}_{i,n-1} = 0$. Then $(A \bmod 2)_{ij} = 1$ iff $\mathcal{P}_{ij} = 1$. All $n/2$ ones of row i lie in columns $0, \ldots, n - 2$. Hence $\mathrm{wt}(\text{row } i) = n/2$. □

Remark 5.2. The column weights of $A \bmod 2$ are *not* constant in general. Column j has weight $|\{i < n-1 : \mathcal{P}_{ij} \neq \mathcal{P}_{i,n-1}\}|$, which depends on the correlation between columns j and $n-1$ of $\mathcal{P}$. This asymmetry arises because the definition of A privileges the last column.

The rows of $\mathcal{P}$ form a constant-weight-$(n/2)$ binary code of length n. The following result shows that a simple combinatorial property of this code—the existence of a pair of rows with disjoint supports—forces a rank deficit.

Proposition 5.3 (Disjoint-support obstruction). *Let $n \equiv 2 \pmod 4$ and let $\mathcal{P} \in \{0,1\}^{n \times n}$ be doubly balanced (all row and column sums equal to $k = n/2$). If $\mathcal{P}$ has two rows with disjoint supports, then $\mathrm{rank}_{\mathbb{F}_2}(\mathcal{P}) \leq n - 1$.*

Proof. Since k is odd, each of the n column sums equals $k \equiv 1 \pmod 2$, so over $\mathbb{F}_2$ the sum of all rows is $\sum_{\ell=1}^{n} r_\ell = \mathbf{1}$. If rows r_i and r_j have disjoint supports of size k each with $2k = n$, then $r_i + r_j = \mathbf{1}$ over $\mathbb{F}_2$. Substituting: $\sum_{\ell \neq i,j} r_\ell = \mathbf{0}$, which is a nontrivial linear dependency among $n - 2 \geq 4$ rows. Hence $\mathrm{rank}_{\mathbb{F}_2}(\mathcal{P}) \leq n - 1$. □

Remark 5.4. The converse of Theorem 5.3 holds at $n = 6$ but fails for $n \geq 10$; see Theorems 7.2 and 7.8 below.

Theorem 5.5 (Odd-prime kernel membership)**.** *Let p be an odd prime dividing n. Then $\mathbf{1} \in \ker(A \bmod p)$, and consequently $p \mid \det(A)$.*

Proof. By Theorem 3.1, $(A\mathbf{1})_i = \frac{n}{2}(n + 1 - 2L_{i,n-1})$. Since $p \mid n$ and $\gcd(p, 2) = 1$, we have $p \mid \frac{n}{2} \cdot (\ldots)$ (indeed $p \mid n$ implies $p \mid (A\mathbf{1})_i$). Thus $A\mathbf{1} \equiv \mathbf{0} \pmod{p}$. By the adjugate identity $\det(A) \cdot \mathbf{1} = \operatorname{adj}(A) \cdot A\mathbf{1}$, it follows that $p \mid \det(A)$. □

Corollary 5.6 (Exact prime structure of the obstruction)**.** *The divisibility $n^2 \mid \det(E_{\mathrm{std}})$ can fail* only *at the prime 2. For all odd prime factors of n, the required divisibility is guaranteed unconditionally. When $n \equiv 2 \pmod{4}$, $n^2 \mid \det(E_{\mathrm{std}})$ holds iff $\operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) < n - 1$ (Theorem 4.3); this can fail for any such n. When $n \equiv 0 \pmod{4}$, $\operatorname{rank}_{\mathbb{F}_2}(A \bmod 2) < n - 1$ holds universally but does not suffice for n^2-divisibility; the complete criterion is $\operatorname{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ (Theorem 4.7).*

Theorem 5.7 (Sufficient condition for $n \equiv 0 \pmod{4}$)**.** *Let n be even. If $\dim \ker_{\mathbb{F}_2}(A \bmod 2) \geq 2$, then $n^2 \mid \det(E_{\mathrm{std}})$.*

Proof. By Theorem 3.2, $h = n/2$ divides $\det(A)$. Write $\det(A) = hq$ with $q \in \mathbb{Z}$. It suffices to show $2 \mid q$, since then $\det(A) \equiv 0 \pmod{2h} = 0 \pmod{n}$ and $\det(E_{\mathrm{std}}) = n \det(A) \equiv 0 \pmod{n^2}$.

By the adjugate identity applied to $\mathbf{1}$:

$$\det(A) \cdot \mathbf{1} = \operatorname{adj}(A) \cdot A\mathbf{1} = h \operatorname{adj}(A) \cdot \mathbf{c},$$

where $c_i = n + 1 - 2L_{i,n-1}$ are all odd (Theorem 3.1). Hence $q = [\operatorname{adj}(A) \cdot \mathbf{c}]_i$ for any i.

Set $B = A \bmod 2$. Since $\dim \ker_{\mathbb{F}_2}(B) \geq 2$, $\operatorname{rank}_{\mathbb{F}_2}(B) \leq n - 3$. Every $(n-2) \times (n-2)$ submatrix of B has rank at most $n-3 < n-2$, so every $(n-2) \times (n-2)$ minor of B vanishes over $\mathbb{F}_2$. Since the entries of $\operatorname{adj}(A)$ are (up to sign) such minors, and $\det(M) \bmod 2 = \det(M \bmod 2)$, we conclude $\operatorname{adj}(A) \equiv 0 \pmod{2}$. Therefore $q = [\operatorname{adj}(A) \cdot \mathbf{c}]_i \equiv 0 \pmod{2}$. □

Remark 5.8. For $n \equiv 0 \pmod{4}$, the condition $\dim \ker_{\mathbb{F}_2}(B) \geq 2$ reduces to: *does* $\ker_{\mathbb{F}_2}(A \bmod 2)$ *contain a vector linearly independent from* $\mathbf{1}$*?* This is sufficient but not necessary: at $n = 8$, approximately 73% of Jacobson–Matthews samples with $\dim \ker = 1$ (the minimum) still satisfy $8 \mid \det(A)$. The complete criterion is given by Theorem 4.7.

6 Smith normal form and p-adic bounds

The adjugate argument in the proof of Theorem 5.7 combines with the Smith normal form to yield a sharper result.

Theorem 6.1 (2-adic lower bound)**.** *Let n be even and let $k = \dim \ker_{\mathbb{F}_2}(A \bmod 2)$. Then*

$$v_2(\det A) \;\geq\; v_2(n/2) + \max(0,\, k - 1),$$

where $v_2(m)$ denotes the 2-adic valuation of m.

Proof. If $\det(A) = 0$, then $v_2(\det A) = +\infty$ by convention, so there is nothing to prove. Hence assume $\det(A) \neq 0$, and set $h = n/2$.

Let $d_1 \mid d_2 \mid \cdots \mid d_{n-1}$ be the invariant factors (Smith normal form) of A over $\mathbb{Z}$. By standard SNF theory [12], the number of invariant factors divisible by 2 equals $\dim \ker_{\mathbb{F}_2}(A \bmod 2) = k$; these are the last k factors $d_{n-k}, \ldots, d_{n-1}$.

Key claim: $v_2(d_{n-1}) \geq v_2(h)$.

Let Δ_{n-2} = gcd of all $(n-2) \times (n-2)$ minors of A (the $(n-2)$-th determinantal divisor), so that $d_{n-1} = \det(A)/\Delta_{n-2}$. The entries of $\operatorname{adj}(A)$ are, up to sign, these $(n-2) \times (n-2)$ minors of A, so $\Delta_{n-2} \mid \operatorname{adj}(A)_{ij}$ for all i, j. Write $\operatorname{adj}(A) = \Delta_{n-2}\, Q$ with $\gcd(Q_{ij}) = 1$.

From the adjugate identity $\det(A) \cdot \mathbf{1} = \operatorname{adj}(A) \cdot A\mathbf{1} = h \operatorname{adj}(A)\, \mathbf{c} = h\, \Delta_{n-2}\, Q\, \mathbf{c}$, we obtain

$$d_{n-1} = \frac{\det(A)}{\Delta_{n-2}} = h\, (Q\, \mathbf{c})_i \quad \text{for each } i,$$

so $h \mid d_{n-1}$ and $v_2(d_{n-1}) \geq v_2(h)$.

Conclusion. If $k = 0$ then $v_2(\det A) \geq v_2(h) + 0$ is just Theorem 3.2. For $k \geq 1$, the k even factors satisfy $v_2(d_{n-k}) \leq \cdots \leq v_2(d_{n-1})$ by the divisibility chain. The last factor contributes $v_2(d_{n-1}) \geq v_2(h)$ and the remaining $k - 1$ factors each contribute $v_2 \geq 1$, giving

$$v_2(\det A) = \sum_{i:\, 2 \mid d_i} v_2(d_i) \;\geq\; v_2(h) + (k - 1). \qquad \square$$

Remark 6.2. The bound of Theorem 6.1 is consistent with every observed kernel dimension at $n = 8, 12, 16$: equality is attained at each value of k realized by Jacobson–Matthews sampling.

The proof above uses nothing specific to $p = 2$: the row-sum identity, the adjugate divisibility, and the SNF pigeonhole all hold at every prime. The only subtlety is that, for n odd, the effective row-sum eigenvalue is $h = n$ rather than $h = n/2$ (Theorem 3.1).

Corollary 6.3 (General p-adic lower bound)**.** *Let p be any prime dividing n, and let $k_p = \dim \ker_{\mathbb{F}_p}(A \bmod p)$.*

- *If n is even, then $v_p(\det A) \;\geq\; v_p(n/2) + \max(0,\, k_p - 1)$.*
- *If n is odd (so p is necessarily odd), then $v_p(\det A) \;\geq\; v_p(n) + \max(0,\, k_p - 1)$.*

For odd p the two cases give the same numerical bound $v_p(n) + \max(0, k_p - 1)$, since $v_p(n/2) = v_p(n)$ when $\gcd(p, 2) = 1$. For every odd prime $p \mid n$, we have $k_p \geq 1$ because $A\mathbf{1} \equiv \mathbf{0} \pmod{p}$ by the row-sum identity, so the odd-prime bound becomes stronger than the base divisibility exactly when $k_p \geq 2$. For $p = 2$, the behavior depends on $n \bmod 4$: if $n \equiv 0 \pmod{4}$, then $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$ by Theorem 4.4; if $n \equiv 2 \pmod{4}$, then $k_2 = 0$ is possible and corresponds to the full-rank obstruction of Theorem 4.3.

Proof. If $\det(A) = 0$, then $v_p(\det A) = +\infty$ by convention, so there is nothing to prove. Hence assume $\det(A) \neq 0$.

The argument follows Theorem 6.1 with 2 replaced by p.

Case n even. By Theorem 3.1, $A\mathbf{1} = h\, \mathbf{c}$ with $h = n/2$ and each c_i odd. The adjugate factorization $d_{n-1} = h\, (Q\mathbf{c})_i$ gives $v_p(d_{n-1}) \geq v_p(h) = v_p(n/2)$. The SNF pigeonhole over the k_p invariant factors divisible by p yields the stated bound.

Case n odd. By Theorem 3.1, $A\mathbf{1} = n\, \mathbf{u}$ with $u_i = (n{+}1{-}2L_{i,n-1})/2 \in \mathbb{Z}$. The adjugate factorization gives $d_{n-1} = n\, (Q\mathbf{u})_i$, so $v_p(d_{n-1}) \geq v_p(n)$. The SNF pigeonhole completes the proof as before. $\square$

Remark 6.4. The bound of Theorem 6.3 is consistent with every observed (n, p, k_p) triple: 2700 Jacobson–Matthews samples across $n \leq 20$ and $p \in \{2, 3, 5\}$ produced zero violations, and equality was observed in every sampled case.

7 Counterexamples

We now exhibit explicit instances of the counterexample mechanisms already identified in Section 4. The aim of this section is illustrative: to show the universal theorem at work in concrete families and to record the finite behavior at the first accessible orders.

In particular, we exhibit Latin squares for which $n^2 \nmid \det(E_{\mathrm{std}})$. For $n \equiv 2 \pmod 4$, Theorem 4.14 guarantees the existence of counterexamples at every order $n \geq 6$; Theorem 4.3 provides the underlying criterion $\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) = n-1$. For $n \equiv 0 \pmod 4$, although $\mathbf{1} \in \ker_{\mathbb{F}_2}(A \bmod 2)$ always holds, the bound $\frac{n^2}{2} \mid \det(E_{\mathrm{std}})$ is still sharp (see $n = 8$ in Theorem 4.4); however, $n^2 \mid \det(E_{\mathrm{std}})$ is guaranteed whenever $\dim \ker_{\mathbb{F}_2}(A \bmod 2) \geq 2$ (Theorem 5.7).

7.1 Exhaustive classification at n=6

There are exactly 9408 reduced Latin squares of order 6 (first row $=$ first column $= (1,2,3,4,5,6)$). Every Latin square of order 6 is equivalent to a reduced one via row and column permutations, which preserve $\det(E_{\mathrm{std}})$ up to sign.

Proposition 7.1. *Among the* 9408 *reduced Latin squares of order* 6*:*

(i) 8832 *satisfy* $36 \mid \det(E_{\mathrm{std}})$*, with* $\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) \leq 4$.

(ii) 576 *satisfy* $18 \mid \det(E_{\mathrm{std}})$ *but* $36 \nmid \det(E_{\mathrm{std}})$*, with* $\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) = 5$ *(full rank).*

The 576 *counterexamples correspond to* 84 *out of* 1252 *distinct parity patterns. No parity pattern gives a mixed outcome: every Latin square with the same pattern produces the same* $\mathbb{F}_2$*-rank.*

The set $\mathcal{R}(6,3)$ of all doubly balanced 6×6 binary matrices has exactly 297 200 elements. We classify these by the action of the group $\mathcal{G} = (S_6 \times S_6) \rtimes \langle \tau, \kappa \rangle$ (row permutations, column permutations, transpose τ, and complement $\kappa : P \mapsto \mathbf{1} - P$), all of which preserve $\mathrm{rank}_{\mathbb{F}_2}$.

Proposition 7.2 (Orbit classification at $n = 6$)**.** $\mathcal{R}(6,3)$ *decomposes into exactly* 6 *orbits under* $\mathcal{G}$*, of sizes* 200*,* 16 200*,* 86 400*,* 43 200*,* 129 600*, and* 21 600*, with* $\mathbb{F}_2$*-ranks* 2*,* 3*,* 4*,* 4*,* 5*,* 6 *respectively. In particular there is a* unique *full-rank orbit. At* $n = 6$*, a doubly balanced pattern has full* $\mathbb{F}_2$*-rank if and only if every pair of rows has nonempty support intersection, i.e. the converse of Theorem 5.3 holds.*

Remark 7.3. The unique full-rank representative admits a block-circulant presentation: indexing rows and columns by $(a,b) \in \mathbb{Z}_3 \times \mathbb{Z}_2$, the matrix is $\mathrm{circ}(J_2, I_2, 0)$ over $\mathbb{Z}_3$. The skip-one circulant $\mathrm{circ}(\{0,1,3\})$ from Theorem 4.14 lies in this orbit.

Remark 7.4. The 84 full-rank parity patterns are characterized by their column weight profiles. Among all $\binom{5}{3}^5 = 100\,000$ binary 5×5 matrices with constant row weight 3, only 19 440 (19.4%) have full $\mathbb{F}_2$-rank, and only 84 of these are realized by Latin squares—a selection rate of 0.43%. The Latin square structure is far more restrictive than the weight constraint alone.

7.2 Constructed counterexamples at n=10

For $n = 10$, exhaustive enumeration is infeasible ($\sim 2 \times 10^{17}$ reduced Latin squares), so we adopt a constructive approach.

Method. By Theorem 4.14, the circulant pattern $\mathcal{P}_{S_{10}}$ with $S_{10} = \{0,1,2,3,5\} \subseteq \mathbb{Z}_{10}$ is doubly balanced and has full $\mathbb{F}_2$-rank. By Theorem 4.12, it lifts to a Latin square. The examples below were found before this theoretical result was available, using the following constructive approach:

1. Generate random *doubly balanced* 10×10 binary matrices $\mathcal{P}$ (5 ones per row and column).
2. Compute the 9×9 matrix $A \bmod 2$ via Theorem 4.2 and check $\mathrm{rank}_{\mathbb{F}_2} = 9$.

3. If full-rank, complete $\mathcal{P}$ to a Latin square by decomposing the bipartite graph into perfect matchings (Theorem 4.12), or alternatively by backtracking with minimum-remaining-values heuristic.

4. Verify $\det(A)$ using exact integer arithmetic (Bareiss algorithm).

Proposition 7.5. (i) *Among* 4000 *doubly balanced* 10×10 *binary matrices sampled via switch-chain MCMC (*50 000 *burn-in steps, thinning every* 200 *steps),* 711 *(*17.8%*) have full* $\mathbb{F}_2$*-rank. The rank distribution is:*

$\mathrm{rank}_{\mathbb{F}_2}$	*6*	*7*	*8*	*9*	*10*
Count	*5*	*107*	*990*	*2187*	*711*

(ii) *All* 711 *full-rank patterns were successfully completed to Latin squares via the perfect-matching decomposition of Theorem 4.12.*

(iii) *Every resulting Latin square had odd* $\det(A)$*, confirming* $\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) = 9$ *and* $100 \nmid \det(E_{\mathrm{std}})$*. Thus the corpus yields* 711 *counterexample instances at* $n = 10$ *(structural distinctness among these instances has not been verified).*

Example 7.6 (Explicit counterexample at $n = 10$)**.** The following is a Latin square of order 10 with $\det(A) = 15\,427\,045$ (odd):

$$L = \begin{pmatrix} 2 & 5 & 6 & 4 & 3 & 8 & 7 & 9 & 10 & 1 \\ 10 & 6 & 2 & 7 & 5 & 1 & 9 & 8 & 4 & 3 \\ 3 & 4 & 1 & 10 & 8 & 5 & 2 & 7 & 9 & 6 \\ 7 & 1 & 10 & 8 & 6 & 9 & 3 & 4 & 2 & 5 \\ 1 & 7 & 3 & 6 & 4 & 10 & 8 & 2 & 5 & 9 \\ 9 & 2 & 5 & 3 & 10 & 6 & 4 & 1 & 8 & 7 \\ 6 & 9 & 8 & 1 & 2 & 7 & 5 & 10 & 3 & 4 \\ 5 & 8 & 7 & 2 & 9 & 4 & 1 & 3 & 6 & 10 \\ 4 & 3 & 9 & 5 & 1 & 2 & 10 & 6 & 7 & 8 \\ 8 & 10 & 4 & 9 & 7 & 3 & 6 & 5 & 1 & 2 \end{pmatrix}$$

The relevant invariants are:

Property	Value
Valid Latin square	✓ (rows and columns are permutations of $\{1,\ldots,10\}$)
Doubly balanced parity	✓ (5 odd + 5 even per row and column)
$\det(A)$	$15\,427\,045 = 5 \times 3\,085\,409$
$v_2(\det(A))$	0 (odd)
$\det(E_{\mathrm{std}}) = 10\det(A)$	$154\,270\,450$
$50 \mid \det(E_{\mathrm{std}})$?	Yes (Theorem 3.2 confirmed)
$100 \mid \det(E_{\mathrm{std}})$?	No $\quad(\Leftarrow$ counterexample)
$\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2)$	$9 = n - 1$ (full rank)

Remark 7.7 (Abundance). The 711 counterexamples at $n = 10$ arise from MCMC-sampled balanced patterns. All satisfy $5 \mid \det(A)$ (as guaranteed by Theorem 5.5) and $2 \nmid \det(A)$. The lift from full-rank balanced pattern to counterexample Latin square succeeded in 711/711 cases (100%), suggesting that the counterexample property depends entirely on the parity pattern (as proved in Theorem 4.11) and that the lift step introduces no additional obstruction.

Remark 7.8 (Converse failure at $n = 10$). Switch-chain MCMC sampling of 20 000 matrices from $\mathcal{R}(10, 5)$ yields approximately 17.8% full-rank. Every full-rank sample satisfies the no-disjoint-pair condition of Theorem 5.3 (recall = 1.000), but the converse fails: among 12 549 samples with no disjoint row pair, 8 995 are rank-deficient (precision = 0.283). Thus the disjoint-support obstruction is necessary but not sufficient for $n \geq 10$.

8 Computational methods

All computations were performed with exact integer arithmetic to avoid numerical artifacts.

All claims explicitly labeled *exhaustive* were verified by finite exact computation. Sampling data are used only as consistency checks or as sources of witnesses, never as proof of a theorem unless exhaustive enumeration is explicitly stated. The scripts used for determinant, Smith normal form, finite-field rank, adjugate testing, and Jacobson–Matthews sampling are part of the accompanying computational project infrastructure.

- **Determinant computation.** The Bareiss algorithm was used to compute $\det(A)$ over $\mathbb{Z}$ in $O(n^3)$ operations, with exact integer division at each step to prevent coefficient growth [1].
- **$\mathbb{F}_2$-rank computation.** Gaussian elimination over $\mathbb{F}_2$ in $O(n^3)$ bit operations.
- **Smith Normal Form.** For selected verification, the Smith Normal Form of A over $\mathbb{Z}$ was computed to extract invariant factors and confirm p-adic valuations independently.
- **Latin square generation.** Reduced Latin squares of order 6 were enumerated exhaustively (9408 squares). For orders 8–16, Latin squares were sampled via the Jacobson–Matthews Markov chain [8] with at least $5n^2$ mixing steps per sample; we refer to these as *Jacobson–Matthews samples* rather than claiming exact uniformity. All 576 Latin squares of order 4 were enumerated for exhaustive verification.
- **Balanced binary matrices.** Random doubly balanced $n \times n$ binary matrices (with row and column sums equal to $n/2$) were generated by rejection sampling for small orders. For $n = 10$, exploratory pattern-space samples were drawn via switch-chain MCMC on $\mathcal{R}(n, n/2)$ (swapping 2×2 checkerboard submatrices to preserve all row and column sums); no rigorous mixing-time bound is claimed, and these samples are treated as approximately uniform.
- **Constrained completion.** Given a doubly balanced parity pattern $\mathcal{P}$, Latin square completion was performed by backtracking with the minimum-remaining-values (MRV) heuristic and forward checking. Each cell's domain is restricted to odd or even values according to $\mathcal{P}$, and assignments are propagated to prune domains in the same row and column.

Remark 8.1. For $n = 6$, the exhaustive enumeration and verification of all 9408 reduced Latin squares completes in under one second. For $n = 10$, the backtracking-lift procedure (pattern generation, rank check, backtracking completion, Bareiss determinant) typically produces a counterexample within several hundred trials, each trial requiring a few seconds.

Code and data availability. All scripts, seeds, and certified datasets used in this paper are available as an open-source Python package (`latin_det`) released under the MIT licence at `https://github.com/aconsciousfractal/Determinant-Divisibility-of-Centered-Latin-Squares`. The repository includes: (i) exact Bareiss determinant, $\mathbb{F}_2$ rank, adjugate and Smith-normal-form routines; (ii) a reproducible random Latin-square sampler and a switch-chain MCMC on $\mathcal{R}(n, n/2)$; (iii) certified datasets for $n \in \{5, 6\}$ exhaustive enumeration, the 10,000-sample $n = 8$ table (seed 0), the 4,000-sample $n = 10$ switch-chain table (seed 42), the cyclic Smith-normal-form table for $n \leq 12$, the p-adic bound scan for

$n \le 12$, $p \in \{2, 3, 5\}$, and the Theorem 7.6 witness JSON certificate; and (iv) a standalone script (`scripts/verify_isotopy_destruction.py`) reproducing the isotopy non-invariance experiment of §9. A SHA-256 manifest (`results/certified/SHA256SUMS`) pins the byte-exact contents of every deterministic artefact. A permanent archive of the submission snapshot (tag `v1.0-submission`) will be deposited on Zenodo; the DOI will be added upon acceptance.

9 Discussion

9.1 Summary of results

Theorem 3.2 provides a universal quadratic divisibility bound for $\det(E_{\mathrm{std}})$ over all Latin squares. Theorem 4.3 reduces the question of improved n^2-divisibility to a binary linear algebra problem when $n \equiv 2 \pmod 4$. Theorem 5.1 and Theorem 5.5 clarify the precise mechanism: every odd prime factor of n contributes its share unconditionally, while the prime 2 is governed by the $\mathbb{F}_2$-rank of a matrix determined solely by the parity pattern of L. For $n \equiv 0 \pmod 4$, the all-ones vector lies automatically in the $\mathbb{F}_2$-kernel of $A \bmod 2$, but this does not suffice to force n^2-divisibility. Theorem 4.7 provides the complete criterion: $n^2 \mid \det(E_{\mathrm{std}})$ if and only if $\mathrm{adj}(B) \cdot \mathbf{1} = \mathbf{0}$ in $\mathbb{F}_2^{n-1}$, where $B = A \bmod 2$; Theorem 5.7 is the special case $\dim \ker_{\mathbb{F}_2} \ge 2$ (which forces $\mathrm{adj}(B) = 0$). Theorem 6.1 refines the 2-adic analysis to a tight lower bound: $v_2(\det A) \ge v_2(n/2) + \max(0, k-1)$ where $k = \dim \ker_{\mathbb{F}_2}(A \bmod 2)$, proved via Smith normal form theory and the divisibility chain of invariant factors. The unified bound $\frac{n^2}{2} \mid \det(E_{\mathrm{std}})$ remains sharp at $n = 8$.

The overall picture is as follows:

n	Divisibility	Status
n odd	$n^2 \mid \det(E_{\mathrm{std}})$	Universal (Theorem 3.2)
n even	$\frac{n^2}{2} \mid \det(E_{\mathrm{std}})$	Universal; globally sharp for even orders, but not at n
$n \equiv 2 \pmod 4$	$n^2 \mid \det(E_{\mathrm{std}})$	iff $\mathrm{rank}_{\mathbb{F}_2}(A \bmod 2) < n-1$ (Theorem 4.3)
$n \equiv 2 \pmod 4$	$n^2 \nmid \det(E_{\mathrm{std}})$	$\exists$ for all $n \ge 6$ (Theorem 4.14)
$n \equiv 0 \pmod 4$	$n^2 \mid \det(E_{\mathrm{std}})$	iff $\mathrm{adj}(A \bmod 2) \cdot \mathbf{1} = \mathbf{0}$ (Theorem 4.7)
n even	$v_2(\det A) \ge v_2(n/2) + \max(0, k-1)$	Lower bound (Theorem 6.1), $k = \dim \ker_{\mathbb{F}_2}$
odd $p \mid n$	$v_p(\det A) \ge v_p(n) + \max(0, k_p-1)$	Lower bound (Theorem 6.3), $k_p = \dim \ker_{\mathbb{F}_p}$

9.2 Consequences and related directions

Isotopy non-invariance. The $\mathbb{F}_2$-rank of $A \bmod 2$—and hence the counterexample property—is *not* an isotopy invariant. Row and column permutations of L preserve $|\det(E_{\mathrm{std}})|$, but symbol relabeling $L \mapsto \gamma(L)$ changes the parity pattern and can alter the $\mathbb{F}_2$-rank; compare the broader isotopy/autotopism context in [13]. In particular, reducing the $n = 10$ counterexample of Theorem 7.6 to standard form (first row $= (1, \dots, 10)$) produces a Latin square with *even* $\det(A)$, i.e. the counterexample property is destroyed. This shows that the divisibility deficit depends on the *labeled* Latin square, not merely on its isotopy class or main class.

Connection to coding theory. By Theorem 5.1, the rows of $A \bmod 2$ form a constant-weight binary code of weight $n/2$ and length $n-1$ (see [10] for the general theory of constant-weight codes). The code is not in general doubly constant-weight: column weights vary depending on the correlation structure of the parity pattern. It would be interesting to characterize which constant-weight configurations arise from Latin squares and how the dimension of the row space is constrained.

The integer invariant $f(L)$. For Latin squares satisfying $n^2 \mid \det(E_{\text{std}})$, the quotient $f(L) = \det(E_{\text{std}})/n^2$ is a well-defined integer. For the cyclic Latin square of odd order n, $f(L) = (-1)^{(n-1)/2} n^{n-3}$. The invariant $f(L)$ is preserved by transposition ($f(L) = f(L^T)$) but not by isotopy in general.

Generalization to odd primes. Theorem 6.3 extends the 2-adic bound of Theorem 6.1 to every prime $p \mid n$: for odd p,

$$v_p(\det A) \;\geq\; v_p(n) + \max(0,\, k_p - 1), \qquad k_p = \dim \ker_{\mathbb{F}_p}(A \bmod p).$$

This strengthens the base divisibility of Theorem 3.2 whenever $k_p \geq 2$. We note that the earlier conjectured bound $v_p(n/p) + \max(0, k_p - 1)$ is valid but sub-optimal: it is weaker by exactly one p-adic unit.

Cokernel and the Latin square graph. Theorem 3.9 shows that the cokernel of the cyclic Latin square has the rigid form $(\mathbb{Z}/n)^{n-2}$. For a general Latin square of order n, the cokernel $K(A)$ exhibits far richer structure: exhaustive enumeration at $n = 5$ yields 6 distinct isomorphism types among 56 reduced squares, and at $n = 6$ there are 377 types among 9408 reduced squares. Related Smith-normal-form computations for combinatorially defined matrices (rook graphs, graph Laplacians) have been carried out in [6, 11] and provide a template for extending the present cokernel analysis.

A natural question is whether $K(A)$ embeds algebraically into the critical group $K(\mathrm{LSG}(n))$ of the associated Latin square graph, which would manifest as divisibility $|K(A)| \mid |K(\mathrm{LSG}(n))|$.

Remark 9.1 (Divisibility by $K(\mathrm{LSG})$). Let $\mathrm{LSG}(n)$ denote the Latin square graph on n^2 vertices (two vertices adjacent iff they share a row, column, or symbol). All Latin square graphs of order n are strongly regular with parameters $(n^2,\, 3(n-1),\, n-2,\, 6)$, hence co-spectral; in particular, $|K(\mathrm{LSG}(n))|$ is constant across all Latin squares of order n (though the group structure may vary). Exhaustive computation gives:

$$|K(\mathrm{LSG}(5))| = 2^{12} \cdot 3^{12} \cdot 5^{22}, \qquad |K(\mathrm{LSG}(6))| = 2^{48} \cdot 3^{53}.$$

The divisibility $|K(A)| \mid |K(\mathrm{LSG}(n))|$ holds for all reduced Latin squares at $n = 3$ and $n = 4$, but *fails* at $n = 5$ for 14 of 56 squares (25%) and at $n = 6$ for 5688 of 7840 non-singular squares (72.6%).

In every observed case, the failure occurs precisely when $|K(A)|$ has a prime factor absent from $|K(\mathrm{LSG}(n))|$. For instance, at $n = 5$, the failing orders $|K(A)| \in \{55, 105, 155\}$ contain the "alien" primes 11, 7, and 31 respectively; at $n = 6$ the prime support of $|K(\mathrm{LSG}(6))|$ is $\{2, 3\}$, and any Latin square with $|K(A)|$ divisible by a prime $p \geq 5$ fails.

9.3 Open questions

(1) **Density of counterexamples.** Several distinct frequencies have been observed. At $n = 6$, the exact counterexample rate among *reduced Latin squares* is $576/9408 \approx 6.1\%$. At $n = 8$, approximately 11% of *Jacobson–Matthews samples* (Latin squares, not patterns) satisfy $n^2 \nmid \det(E_{\text{std}})$ (Theorem 3.7). At $n = 10$, the doubly balanced *parity-pattern* corpus of Theorem 7.5 yields $711/4000 \approx 17.8\%$ full-rank patterns. These three rates are not directly comparable: the first is exact over reduced squares, the second is a Latin-square sampling frequency, and the third is a pattern-space frequency. Clarifying the relationship among these measures, and determining the asymptotic counterexample density among all Latin squares as $n \to \infty$ along even n, remains open.

(2) **Structural characterization of full-rank parity patterns.** At $n = 6$, Theorem 7.2 shows that full $\mathbb{F}_2$-rank occurs precisely for the unique orbit of doubly balanced patterns

with no disjoint row pair; the converse of Theorem 5.3 holds exactly. For $n \geq 10$ with $n \equiv 2 \pmod 4$, the disjoint-pair obstruction remains necessary but is no longer sufficient (Theorem 7.8). Find a structural characterization of full-rank parity patterns—readable directly from the binary matrix—that extends beyond $n = 6$.

(3) **Group-theoretic structure.** Among the 80 reduced Latin squares of order 6 that are Cayley tables of groups, 36 (45%) are counterexamples—a much higher rate than the 6.1% among all reduced Latin squares. What role does the group structure play in determining the $\mathbb{F}_2$-rank?

(4) **Prime support of $K(\mathrm{LSG}(n))$.** By Theorem 9.1, the prime support of $|K(\mathrm{LSG}(n))|$ is $\{2,3,5\}$ at $n = 5$ and $\{2,3\}$ at $n = 6$. Does the prime support grow or shrink with n? A complete description of $|K(\mathrm{LSG}(n))|$ would determine exactly which cokernels $K(A)$ can embed into the graph-level critical group.

A Determinant values at n=5

For all 56 reduced Latin squares of order 5, $\det(E_{\mathrm{std}})$ is divisible by $25 = 5^2$. The distribution of values is:

$\det(E_{\mathrm{std}})$	Count
−525	6
−400	12
−25	16
25	5
275	2
400	8
525	4
625	1
775	2

In particular, $\det(E_{\mathrm{std}}) \neq 0$ for all 56 squares, so $\mathrm{rank}(E_{\mathrm{std}}) = 4 = n - 1$ (full rank on V_{std}) is universal at $n = 5$.

B Weight structure of n=10 counterexamples

A representative sample of eight counterexamples at $n = 10$, drawn from the corpus of Theorem 7.5, exhibits constant row weight $5 = n/2$ in $A \bmod 2$ (as predicted by Theorem 5.1) and variable column weights:

#	$\det(A)$	Column weights of $A \bmod 2$ (sorted)
1	2 606 485	$(1, 4, 4, 5, 5, 6, 6, 7, 7)$
2	−8 863 025	$(3, 3, 4, 5, 5, 6, 6, 6, 7)$
3	−25 279 735	$(2, 4, 4, 5, 5, 5, 6, 7, 7)$
4	−24 821 965	$(2, 2, 4, 5, 5, 6, 7, 7, 7)$
5	−50 014 675	$(2, 4, 4, 5, 5, 5, 6, 7, 7)$
6	42 371 805	$(3, 4, 4, 5, 5, 5, 6, 6, 7)$
7	−11 745 775	$(1, 5, 5, 5, 5, 6, 6, 6, 6)$
8	5 123 485	$(2, 4, 4, 5, 5, 5, 6, 7, 7)$

All 711 corpus determinants are divisible by 5 (consistent with Theorem 5.5) and none is divisible by 2.

C Exhaustive column weight analysis at n=6

Among the 576 counterexamples at $n = 6$, the column weight distributions of A mod 2 are:

Column weights (sorted)	Count
$(2, 3, 3, 3, 4)$	320
$(1, 3, 3, 4, 4)$	256

All 576 counterexamples have constant row weight $3 = n/2$.

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